

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

15th March 2017

**Subject CA1 – Actuarial Risk Management
(Paper II)**

Time allowed: 3 Hours (14.45* - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *Attempt all questions, beginning your answer to each question on a separate sheet.*
3. **You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have 3 hours to complete the paper.*
4. *You must not start writing your answers until instructed to do so by the Supervisor.*
5. *Mark allocations are shown in brackets.*
6. *Please check if you have received complete Question Paper and no page is missing. If so kindly get new set of Question Paper from the Invigilator*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** i) Explain why certain health insurance products are contracts of indemnity while others provide pre-specified amount on certain contingencies. (3)
- ii) Why an insurance company's approach to financial underwriting may differ for these two types of products. (2)
[5]
- Q. 2)** "Defined Contribution retirement benefit schemes pass all the risks to its members". Discuss the validity of this statement. [8]
- Q. 3)** A large financing company is planning to set up a general insurance subsidiary. Initially the company plans to sell comprehensive motor insurance policies to its vehicle loanees.
- i) List the main users of data in a general insurance company? (3)
- ii) Outline the principal factors the housing company should take into account while designing a system for administering this business. (5)
- iii) Outline how the company can use its existing data in designing and pricing the motor product. (3)
[11]
- Q. 4)** A life insurance company, mainly selling unit linked contracts, is planning to design a new unit linked critical illness product. Under the product, benefit is payable on maturity, earlier death/withdrawal or earlier diagnosis of any of 15 listed critical illnesses. The benefit on diagnosis of any of the listed critical illness is higher of the unit linked fund value or a guaranteed sum insured. On maturity or earlier death/withdrawal the benefit is the fund value.
- Describe how you will estimate the following assumptions for pricing the above product:
- i) Critical illness rate (5)
- ii) Expenses (5)
- iii) Interest rate (3)
[13]
- Q. 5)** A company is planning to launch a cab company in a city. People would be able book a cab using the company's mobile application.
- i) Briefly explain the likely set up costs and on-going costs for the cab company? (5)
- ii) Outline the major risks the company faces and the ways it can be mitigated. (10)
[15]

Q. 6) A proprietary life insurance company has a large portfolio of without profit immediate annuity contracts. The regulation in the country requires the investments backing the annuity liabilities to be in Government securities. Now the company is in the process of doing its annual regulatory valuation. In the past one year the following events have happened.

- Successful Clinical trials with a new drug, that could prolong and improve quality of life of people, suffering from a certain illness and which till date was thought to be terminal. The illness primarily affects those above age 50.
- The yield on Government securities have increased by 10%.
- The company has introduced a system of direct bank account transfers for payment of annuities as a result of which its overall expenses for this business has reduced by 10% compared to the previous year.

- i)** Explain the factors to be taken into account while finalizing the assumptions for the current year valuation. (5)
 - ii)** Discuss the impact of these changes on the surplus arising during the year. (5)
 - iii)** Outline the other areas of actuarial review which would need to be conducted as a result of the above events. (5)
- [15]**

Q. 7) The trustees of a large final salary pension scheme are considering allowing transfer-ins, i.e. employees bringing in funds from their previous pension arrangements.

Their Consultant actuary has suggested two options:

- Option 1: Using the funds transferred in, purchase an equivalent additional benefit in the final salary scheme
- Option 2: Have a separate accumulation account for such transfer in funds and maintain member level fund values, used to purchase annuity at retirement.

Outline the design factors the trustees should consider when choosing between the two options

[16]

Q. 8) **i)** Outline the factors an actuary should take into account while doing a professional job. (5)

An actuary is advising an insurance company in setting the discontinuance terms for regular premium without profit endowment assurance contracts.

- ii)** Identify the principal stakeholders and their aims/interests (4)
 - iii)** The marketing manager has suggested that to make the product more attractive, the insurance company should give a minimum of 70% of the premiums paid as surrender value, in case of surrender within the first three policy years. How would you evaluate the viability of this suggestion? (8)
- [17]**
